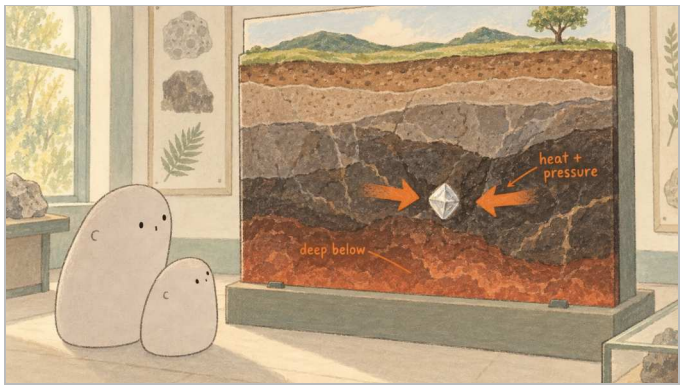
The flash that wouldn't stay.

01 Could a diamond flash where no light can reach it?



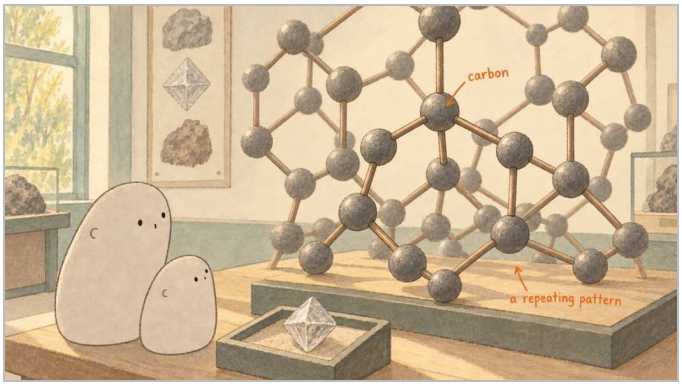
An imagined window into the diamond's dark beginning. The crystal is enlarged; layers and distances are simplified.

Look closer. Find the tiny landscape, the crystal, and the two orange arrows. What looks bright? What looks dark?

A little explanation. Natural diamonds grew from carbon deep underground, where it was very hot and the pressure was enormous. With no light reaching it, a diamond there could not send a flash to anyone's eyes.

Try this. Cover the crystal in the picture with one finger, then uncover it. Tell your grown-up where you would search next for the flash.

02 Could the diamond's tiny pattern be making the flash?



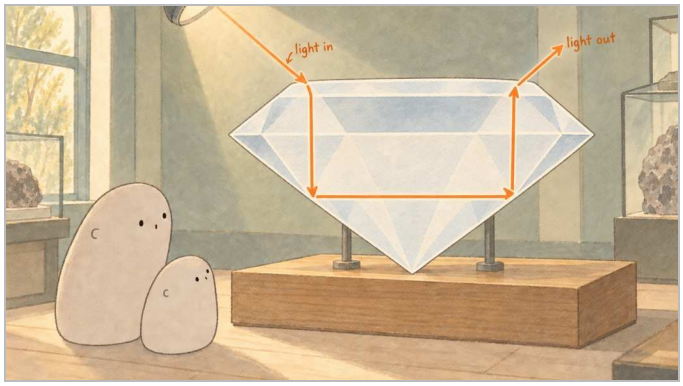
The model reveals a repeating carbon pattern—not a hidden source of light. Real atoms are much too small to see.

Look closer. Look from the small stone to the large model. Follow one rod to the next. Do you see a lamp or another source of light?

A little explanation. A diamond is made mostly of carbon atoms joined in a repeating pattern. The balls and rods help us picture parts far too small to see, but the pattern is not a hidden lamp.

Try this. Trace three connections with your finger. Say what this model explains—and what it still does not explain.

03 Which thing in this picture starts the bright path?



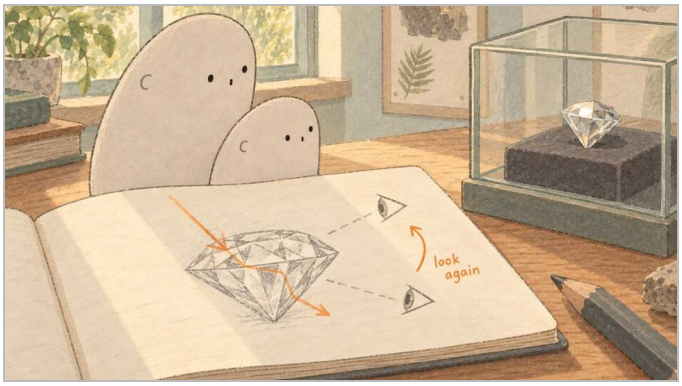
The lamp begins this simplified path. Cut facets change the light's direction and send some of it back out.

Look closer. Begin at the orange line. Follow it backward until you cannot go any farther. Where did you land?

A little explanation. The path begins at the lamp. Light bends when it enters the diamond and can bounce inside. Carefully cut facets help send some of that borrowed light back toward our eyes.

Try this. Trace the whole path from lamp to diamond to eye. Imagine the lamp switched off. What part of the path would disappear?

04 Why might the flash move when Little moves?



Little's two viewing spots turn a quick flash into an observation they can keep.

Look closer. Find the two eye positions in the sketch. Follow each line toward the drawn diamond. What changed?

A little explanation. A flash reaches our eyes only along certain paths. When the light, diamond, or viewer moves, a bright facet can dim while another one flashes.

Try this. Point to one eye position, then the other. Pretend you are Little and tell where you would look next.